Continental Drift

My Note Sheet · Earth Science

Name _____

Date _____ Hour _____

Big Question: Was Alfred Wegener right that the continents moved?
You are a scientist. Use clues (evidence) to decide.

1 Words to know

continent
a very large piece of land
my language: _____

ocean
a very large body of salt water
my language: _____

coastline
where the land meets the ocean
my language: _____

fossil
the remains or print of a plant or animal from long ago
my language: _____

mountain chain
a long line of mountains
my language: _____

evidence
clues that help show something is true
my language: _____

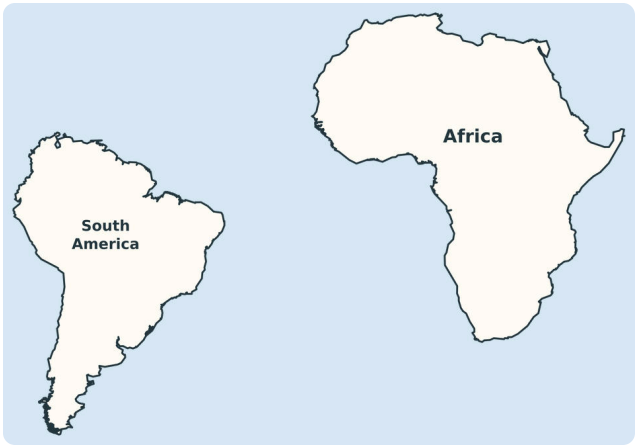
Pangaea
the one giant piece of land long ago
my language: _____

drift
to move slowly
my language: _____

hypothesis
a smart guess you can test
my language: _____

2 The big idea: the continents were once joined

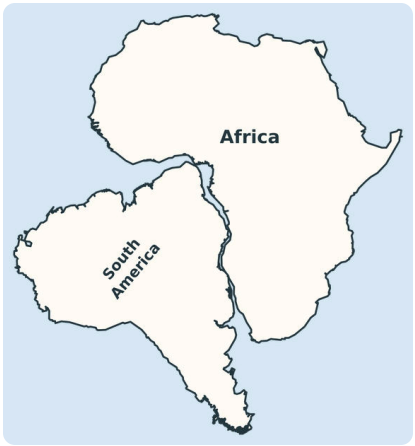
TODAY



The continents are far apart.
The Atlantic Ocean is between them.

go back in time

LONG AGO (Pangaea)



The coastlines fit like puzzle pieces.
South America turns to fit Africa.

Alfred Wegener said: Long ago, all the continents were joined in **Pangaea**. Then they **drifted** apart. They still move today, about as fast as your **fingernails grow**.

3 Wegener's story

1912
Wegener shares his big idea.

People say "No!"
"How can a whole continent move?"

1930
Wegener dies. He never found the answer.

Later
Scientists find new clues in the ocean.

My notes What do I notice? What questions do I have?

4 Four kinds of clues (evidence)

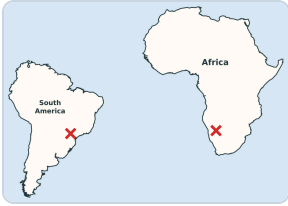
Coastlines



Found: Africa and South America have matching coasts.

So: the land was once together.

Fossils



Mesosaurus (small reptile) lived in lakes and rivers. It could not cross a salty ocean. Its fossils (**X**) are on **both** continents.

Glossopteris plant fossils are on many continents.

So: the land was joined.

Mountain chains

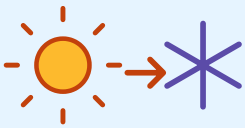


Mountains in southern Africa match mountains in southern South America.

The **Appalachians** (North America) match mountains in **Scotland**.

So: one chain was split by an ocean.

Rocks and glaciers



Glacier scratches are in warm places, like **India** and Africa.

The **same rock types** are on both sides of the ocean.

A glacier needs a cold place.

So: these lands were once in a cold place.

5 Wegener's problem

He HAD this (evidence)

- ✓ Coastlines that fit
- ✓ The same fossils on two continents
- ✓ Matching mountain chains
- ✓ Glacier marks in warm places

He did NOT have this

- ? An answer for **how** continents move
- ? A force strong enough to move land
- ? A way to measure the moving land
- ? Clues from the ocean floor

6 My hypothesis

A hypothesis is a smart guess. What could push or pull a continent? There is no wrong answer.

I think the continents move because

_____

_____

7 Write a strong answer: Claim · Evidence · Reasoning

Claim

What you think is true.

I think Wegener was (right / wrong) because _____.

Evidence

A clue. Say **where** it was found.

One clue is _____. It was found on _____ and _____.

Reasoning

Explain **why** the clue proves your claim.

This shows the continents moved because _____.

Example: I think Wegener was right. Glossopteris plant fossils are on Africa, South America, India, Australia and Antarctica. A plant cannot cross a big ocean, so these lands were once joined.

Tip: use at least 2 clues. Say where each clue was found.